Karlsruher Institut für Technologie

KIT Karlsruhe Institute of Technology
KDD Workshop on Evaluation and Trustworthiness of Agentic AI (ETAAI'26) | Jeju, Korea

Governed LLM-driven ML experimentation

The LLM proposes; the control plane validates and adjudicates; the ledger makes every trial auditable.

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1. Task scores evaluate results, not experimental conduct

What must hold → what we measure

A higher AUC never shows whether the agent stayed in scope or who decided acceptance. Each condition becomes a reported metric.

D1 Bounded edit scope → invalid-action rate
No protected benchmark, data or evaluation file changes.

D2 Complete attempt accounting → failure taxonomy
Every attempt recorded — invalid proposals and crashes.

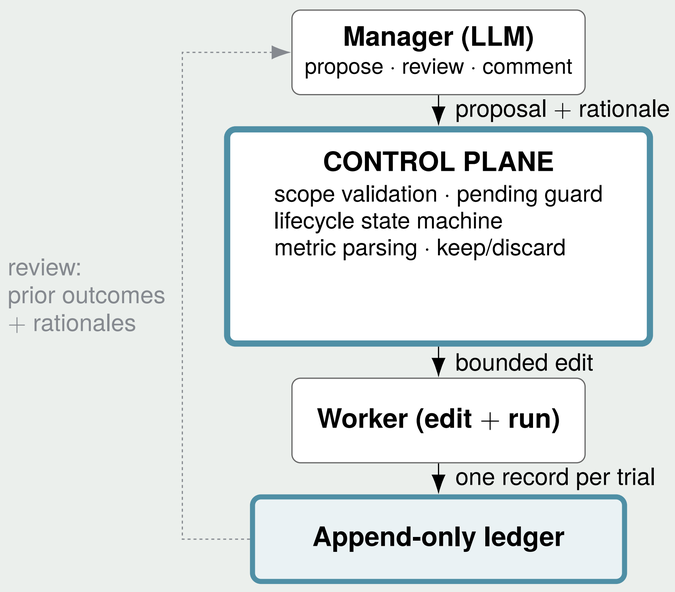
D3 Independent adjudication → acceptance rate
Fixed control-plane rules decide, never the agent.

D4 Provenance linkage → provenance completeness
Each attempt links to patch, config, run, metric, decision.

D5 Recomputability → artifact manifest
Claims recompute from the ledger; no training rerun.

2. The governed control plane

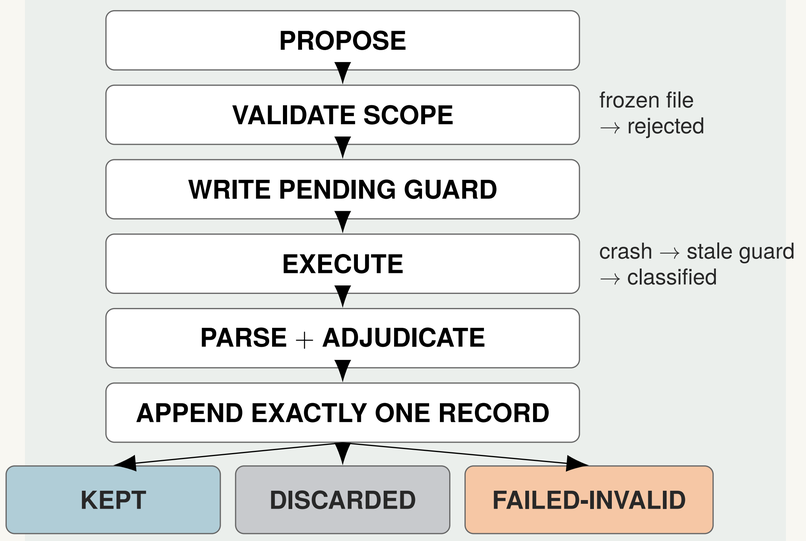
Agent = Model + Harness



Memory is an optional review channel; the ledger is the authoritative audit record — owned by the control plane. Managers and workers are swappable; the contract is fixed.

3. The governed trial contract

Exactly one immutable record per trial



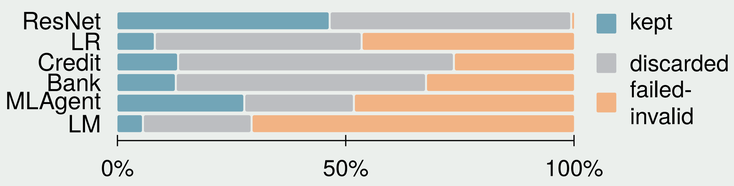
Every opened budget slot appends exactly one terminal record before the next may open.

Structural, not empirical: the invariant holds by construction.

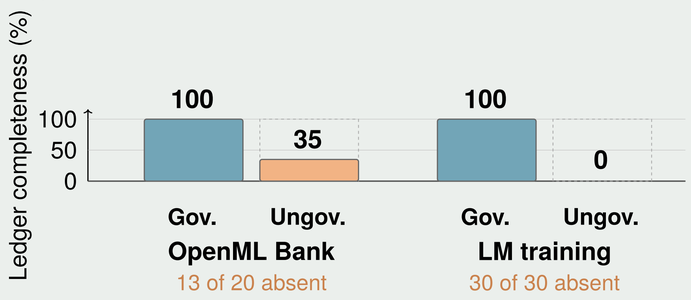
4. What the evidence shows

6 nodes · 1,445 trials · 100% provenance

One contract; only the NodeSpec YAML differs between nodes.



Acceptance 0–47%, invalid 0.4–100%.



43 of 50 ungoverned trials left no ledger record; 0 of 50 governed.

5. Next: predicting failure

R1 — failure-dynamics benchmark

The audit record shows *that* a trial failed — not *when* the trajectory started to go wrong.

Step-granular contract
Typed per-step events: state differences, evidence provenance, memory reads.

Causal fault injection
Seven mechanism families, deterministic injectors, a clean counterfactual per fault.

Calibrated early warning
$q_t = P(\text{terminal failure} \mid H_t)$ and a cause distribution.
Reported: hazard, propagation, lead time, recoverability window.

No replicated failure law and no warning gain ⇒ released as benchmark plus negative result.

6. Next: acting on the warning

R2 — cost-sensitive verified recovery

A warning still does not say whether evidence is worth buying, or which recovery is safe.

Belief in, action out
From frozen R1: failure hazard, cause, evidence freshness, reversibility, damage.

Action space
continue · verify · recover (retry / repair / rollback / re-plan) · escalate · abstain

Matched-budget comparison
Against never, always, periodic, random and confidence-threshold verification.

Process supervision
Train only on independently adjudicated recovery transitions.

Same rule throughout: nothing enters the record unless an authority other than the agent verified it.

Paper + code + archived ledgers
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Refs: [1] Trivedy *et al.* (2026); [2] Zaharia *et al.* (2024).
All governance metrics recomputable from the ledgers — no GPU required.